计算机体系结构
实验 3
例外与中断的添加

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1 实验目标

本实验接着实验 2 的五级流水线，完成《CPU 设计实战》第 7 章两个实践任务。CPU 仍是 `mycpu_verify/rtl/myCPU`。

1. 增加 `MFC0`、`MTC0`、`ERET` 和 `SYSCALL`，以及 CP0 的 Status、Cause、EPC，使 `func_lab8` 的 69 个功能点与参考轨迹一致。
2. 增加 Break、地址错、整数溢出和保留指令，以及 Count、Compare、BadVAddr。定时器中断固定在硬件中断 5，即 Cause 的 IP7。硬件中断 0～5 对应 IP7～IP2，软件中断对应 IP1、IP0。`func_lab9` 共 94 个功能点。

前递、load-use 停顿、HI/LO 和延迟槽沿用实验 1、实验 2。没有实验箱，比特流和记忆游戏不上板，结论以 trace 对比为准。

2 实验原理

2.1 CP0 与例外入口

CP0 放在写回级。复位时 Status 为 `0x10400004`，即 CU0、BEV、ERL 为 1；Cause 的 ExcCode 为 `0x1f`。BEV 为 1 时入口是 `0xbfc00380`，否则是 `0x80000180`。`func_lab8` 先执行 `mtc0`，把 Status 写成 `0x00400000`，BEV 仍为 1，ERL 被清掉，随后的 `SYSCALL` 进入 `0xbfc00380`。

`MTC0` 写 Status 时只改 CU0、BEV、NMI、IM、KSU、ERL、EXL、IE。例外把 EXL 置 1。`ERET` 在 ERL 为 0 时清 EXL，并清 ERL。EPC 和 Cause.BD 只在 EXL 为 0 时更新：受害指令在延迟槽中时 EPC 为该指令 PC 减 4，否则 EPC 就是该指令的 PC。Cause 的域顺序与参考 CPU 相同，测试程序随后会用掩码判断 ExcCode，trace 则比较 `MFC0` 写出的整字。

例外	ExcCode
中断	0
取指或读地址错（AdEL）	4
写地址错（AdES）	5
`SYSCALL`	8
`BREAK`	9
保留指令	10
整数溢出	12

ExcCode 0 是合法的中断编码，所以流水线另用一位表示“本条是例外”，写回级再决定把它当成中断还是这条指令自己的例外。

2.2 精确例外

SYSCALL、BREAK、保留指令和取指地址错在译码级标出。溢出、LW/LH/LHU 的地址错和 SW/SH 的地址错要等执行级算出地址或加减结果。例外指令一旦进入执行级，ex_cancel 清掉更年轻的取指和译码，并挡住新指令进入执行级。因此 SYSCALL 后面的 sw、mtc0 或乘除不会改内存、EPC 或 HI/LO。写回级才更新 CP0，并向取指发出冲刷：ERET 去 EPC，其余例外去入口。

指令 RAM 的读数据晚一拍。冲刷的当拍就用入口或 EPC 发读请求，同时把取指级的 PC 和有效位置上。下一拍返回的指令和这个 PC 一起进入译码。

2.3 延迟槽与返回

译码级停着分支或跳转时，取指级正在取的那一条是延迟槽，BD 随它往后流。延迟槽里发生例外时 EPC 记分支的 PC。通用处理程序读 Cause.BD：为 0 则把 EPC 加 4，否则加 8，再 MTC0 写回并 ERET。这样延迟槽例外返回到分支后面第二条，槽里的指令不会再执行一次。

中断处理程序不看 BD，直接把 EPC 加 8。定时器和软件中断的测试在 1: b 1b; nop 上自旋，并要求 EPC 等于标号 1。受害指令若是这条分支，EPC 就是它的 PC；若是槽里的 nop，BD 为 1，EPC 仍是分支的 PC。两种情况返回地址都是分支加 8。该处理程序在读 Cause 时关掉 trace 比较，因为 BD 可能不同。

2.4 中断

```
int_req = IE && !EXL && !ERL && |(IM & IP)
```

中断在写回级受理，受害指令是当时写回的那一条，并且它本身不是例外。受理后 EXL 置 1，后面的指令被冲刷，处理程序里不再嵌套。

Count 用一位每拍翻转的使能，隔拍加 1，与参考 CPU 的 count_add_en 一致。Count 等于 Compare 时 TI 置 1，MTC0 写 Compare 时 TI 清 0。IP7～IP2 比中断源晚一拍：IP7 是硬件中断 5 与 TI 的或，IP6～IP2 是硬件中断 4～0，IP1 和 IP0 由 MTC0 写 Cause 的对应位。SoC 把 6 位硬件中断输入接成 0。func_lab9 的定时器只靠 TI，软件中断只写 IP1/IP0。

2.5 MFC0 的数据相关

MFC0 的结果在写回级才从 CP0 读出，执行级和访存级没有这个数。它停在这两级、并且目的寄存器正是译码级要读的寄存器时，译码级停住，前递也不把 ALU 的值送出去。它到达写回后，写回数据换成 CP0 的读数，实验 1 的写回前递继续可用。处理程序里紧挨着的是 mfc0 k0; bne k0, ...，这一拍停顿是必需的。

3 实验过程

3.1 程序与轨迹

func_lab8、func_lab9 放在 CPU_CDE/soft/，与 func_lab7 同级。两个工程里的 inst_ram 都是锁定的旧 IP，先 upgrade_ip，再把 COE 指到对应的 obj/inst_ram.coe 并重新生成仿真文件。cpu132_gettrace 跑完得到 golden_trace.txt，mycpu 的 testbench 在写非 0 寄存器时逐条比较。

func_lab8 的测试数为 $23 + 19 + 26 + 1 = 69$，新增的一点是 SYSCALL。func_lab9 再加 25 点，共 94。新增点依次是 Break，ADD/ADDI/SUB 溢出，LW/LH/LHU 读地址错，SW/SH 写地址错，取指地址错，保留指令，定时器中断，软件中断，以及放在 BEQ 到 JALR 延迟槽里的例外。

3.2 func_lab8

69 个功能点一次通过。第 42 点 mycpu 为 1077305 ns，gettrace 为 1193795 ns。第 68 点分别为 1644675 ns 和 1835425 ns。这两个时刻与实验 2 的 func_lab6、func_lab7 相同。

第 69 点是 SYSCALL：mycpu 1649965 ns，gettrace 1841065 ns。结束时间见第 5.1 节。

3.3 func_lab9

换成 func_lab9 的 COE 后，前 69 点的通过时刻与 func_lab8 相同，第 70 到 94 点继续通过。第 94 点 mycpu 为 1939805 ns，gettrace 为 2130615 ns。两边从第 69 点到第 94 点都增加约 0.290 ms。

4 关键代码

4.1 译码级的例外

取指 PC 的低两位非 0 时是 AdEL，BadVAddr 就是这个 PC。操作码不属于实验 2 已实现的指令，也不是 MFC0/MTC0/ERET/ SYSCALL/BREAK 时，记为保留指令。ADD、ADDI、SUB 只先带上溢出检查使能，真正的判断在执行级。

```
wire inst_adel_if = |ds_pc[1:0];
wire inst_ri      = ~inst_known & ~inst_adel_if;
wire ds_ex        = inst_syscall | inst_break | inst_ri | inst_adel_if;
wire [4:0] ds_excode = inst_adel_if ? `EXC_ADEL :
                       inst_syscall ? `EXC_SYS  :
                       inst_break   ? `EXC_BP   :
                       inst_ri      ? `EXC_RI   : 5'd0;
wire ds_ov_en = inst_add | inst_addi | inst_sub;
```

这类指令的访存操作和乘除操作改成空，避免保留指令的位型撞上 DIV 时把执行级停住。

4.2 执行级

有符号加减在两操作数符号相同（减法为相异）、结果符号与被减数或被加数不同时溢出。ADDU/ADDIU/SUBU 不带 es_ov_en。字地址错看最低两位，半字地址错看最低一位。例外指令的写使能为 0。

```
wire es_ov = es_ov_en && (es_alu_op[1]
             ? (es_alu_src1[31] != es_alu_src2[31])
               && (alu_result[31] != es_alu_src1[31])
             : (es_alu_src1[31] == es_alu_src2[31])
               && (alu_result[31] != es_alu_src1[31]));
wire es_ex_f   = es_ex || es_ov || es_adel_m || es_ades_m;
assign ex_cancel = es_valid && (es_ex_f || es_cp0_op == `CP0_ERET);
assign data_sram_wen = (es_valid && !wb_flush && !es_ex_f) ? st_wen : 4'h0;
```

地址错时 BadVAddr 取 ALU 结果。取指地址错的 BadVAddr 仍是译码级传来的 PC。

4.3 CP0

例外优先于 MTC0 和 ERET 修改 EXL。EPC 在 EXL 已为 1 时保持不变。

```
if (ws_valid && ws_mtc0 && ws_addr == 5'd11)
    cause_ti <= 1'b0;
else if (count == compare)
    cause_ti <= 1'b1;

cause_hwip <= {hw_int[5] | cause_ti, hw_int[4:0]};

if (ws_valid && ws_ex && !status_exl)
    cause_epc <= ws_bd ? (ws_pc - 32'd4) : ws_pc;
if (ws_valid && ws_ex)
    status[1] <= 1'b1;
else if (ws_valid && ws_eret)
    status[1] <= status_erl ? status[1] : 1'b0;
```

MFC0 按寄存器号读 BadVAddr、Count、Compare、Status、Cause、EPC，其余号返回 0。

4.4 冲刷与 MFC0 停顿

```
assign inst_sram_addr = wb_flush ? wb_flush_pc : nextpc;
else if (wb_flush) begin
    fs_valid <= 1'b1;
```

```
    fs_pc    <= wb_flush_pc;
end

wire mfc_use = (es_mfc && (es_dest != 5'd0) && (...))
            || (ms_mfc && (ms_dest != 5'd0) && (...));
assign ds_ready_go = ~load_use && ~mfc_use;
assign rf_we    = ws_gr_we && ws_valid && !take_ex;
assign rf_wdata = ws_mfc0 ? cp0_rdata : ws_final_result;
```

例外指令即使译码时 `gr_we` 为 1（溢出的 `ADD`、地址错的 `LW`），写回级也不写通用寄存器。中断的受害指令仍按原意写回，EPC 指向它，返回后会再执行。

5 实验结果

5.1 功能点对比

仿真控制台每隔约 10 µs 打印一行 `Test is running`。下面保留功能点通过行和结束行。左栏是 `cpu132_gettrace`，右栏是 `mycpu`。`gettrace` 结束时打印 `gettrace end!`；`mycpu` 结束时打印 `----PASS!!!`。

`func_lab8` 两边都到第 69 点。`gettrace` 在 1841525 ns 结束，`mycpu` 在 1650425 ns 结束。前 68 点与实验 2 的 `func_lab7` 时刻相同。

gettrace

```
Test begin!
----[  14055 ns] Number 8'd01 Functional Test Point PASS!!!
----[  41545 ns] Number 8'd02 Functional Test Point PASS!!!
----[  63015 ns] Number 8'd03 Functional Test Point PASS!!!
----[  90505 ns] Number 8'd04 Functional Test Point PASS!!!
----[ 117995 ns] Number 8'd05 Functional Test Point PASS!!!
----[ 145485 ns] Number 8'd06 Functional Test Point PASS!!!
----[ 172975 ns] Number 8'd07 Functional Test Point PASS!!!
----[ 200465 ns] Number 8'd08 Functional Test Point PASS!!!
----[ 227955 ns] Number 8'd09 Functional Test Point PASS!!!
----[ 255445 ns] Number 8'd10 Functional Test Point PASS!!!
----[ 276915 ns] Number 8'd11 Functional Test Point PASS!!!
----[ 298385 ns] Number 8'd12 Functional Test Point PASS!!!
----[ 319855 ns] Number 8'd13 Functional Test Point PASS!!!
----[ 333695 ns] Number 8'd14 Functional Test Point PASS!!!
----[ 348585 ns] Number 8'd15 Functional Test Point PASS!!!
----[ 420255 ns] Number 8'd16 Functional Test Point PASS!!!
----[ 496875 ns] Number 8'd17 Functional Test Point PASS!!!
----[ 559275 ns] Number 8'd18 Functional Test Point PASS!!!
----[ 613675 ns] Number 8'd19 Functional Test Point PASS!!!
----[ 615745 ns] Number 8'd20 Functional Test Point PASS!!!
----[ 617815 ns] Number 8'd21 Functional Test Point PASS!!!
----[ 619425 ns] Number 8'd22 Functional Test Point PASS!!!
----[ 621035 ns] Number 8'd23 Functional Test Point PASS!!!
----[ 643935 ns] Number 8'd24 Functional Test Point PASS!!!
----[ 665405 ns] Number 8'd25 Functional Test Point PASS!!!
----[ 687855 ns] Number 8'd26 Functional Test Point PASS!!!
----[ 709325 ns] Number 8'd27 Functional Test Point PASS!!!
----[ 730795 ns] Number 8'd28 Functional Test Point PASS!!!
----[ 752265 ns] Number 8'd29 Functional Test Point PASS!!!
----[ 773735 ns] Number 8'd30 Functional Test Point PASS!!!
----[ 795205 ns] Number 8'd31 Functional Test Point PASS!!!
----[ 822695 ns] Number 8'd32 Functional Test Point PASS!!!
----[ 850185 ns] Number 8'd33 Functional Test Point PASS!!!
----[ 877675 ns] Number 8'd34 Functional Test Point PASS!!!
----[ 935355 ns] Number 8'd35 Functional Test Point PASS!!!
----[1019275 ns] Number 8'd36 Functional Test Point PASS!!!
----[1093745 ns] Number 8'd37 Functional Test Point PASS!!!
----[1160535 ns] Number 8'd38 Functional Test Point PASS!!!
----[1170115 ns] Number 8'd39 Functional Test Point PASS!!!
----[1179675 ns] Number 8'd40 Functional Test Point PASS!!!
----[1188615 ns] Number 8'd41 Functional Test Point PASS!!!
----[1193795 ns] Number 8'd42 Functional Test Point PASS!!!
----[1222025 ns] Number 8'd43 Functional Test Point PASS!!!
----[1265245 ns] Number 8'd44 Functional Test Point PASS!!!
----[1308185 ns] Number 8'd45 Functional Test Point PASS!!!
----[1351405 ns] Number 8'd46 Functional Test Point PASS!!!
----[1385005 ns] Number 8'd47 Functional Test Point PASS!!!
----[1432805 ns] Number 8'd48 Functional Test Point PASS!!!
----[1482775 ns] Number 8'd49 Functional Test Point PASS!!!
----[1537775 ns] Number 8'd50 Functional Test Point PASS!!!
----[1539745 ns] Number 8'd51 Functional Test Point PASS!!!
----[1547845 ns] Number 8'd52 Functional Test Point PASS!!!
----[1556585 ns] Number 8'd53 Functional Test Point PASS!!!
----[1565005 ns] Number 8'd54 Functional Test Point PASS!!!
----[1573745 ns] Number 8'd55 Functional Test Point PASS!!!
----[1583125 ns] Number 8'd56 Functional Test Point PASS!!!
----[1591865 ns] Number 8'd57 Functional Test Point PASS!!!
----[1598255 ns] Number 8'd58 Functional Test Point PASS!!!
----[1627845 ns] Number 8'd59 Functional Test Point PASS!!!
----[1659605 ns] Number 8'd60 Functional Test Point PASS!!!
----[1682825 ns] Number 8'd61 Functional Test Point PASS!!!
----[1709545 ns] Number 8'd62 Functional Test Point PASS!!!
----[1724385 ns] Number 8'd63 Functional Test Point PASS!!!
----[1740065 ns] Number 8'd64 Functional Test Point PASS!!!
----[1771905 ns] Number 8'd65 Functional Test Point PASS!!!
----[1801825 ns] Number 8'd66 Functional Test Point PASS!!!
----[1818865 ns] Number 8'd67 Functional Test Point PASS!!!
----[1835425 ns] Number 8'd68 Functional Test Point PASS!!!
----[1841065 ns] Number 8'd69 Functional Test Point PASS!!!
============================================================
gettrace end!
----Succeed in generating trace file!
$finish called at time : 1841525 ns : File "D:/Architecture/lab1/
↪  lab3/CPU_CDE/cpu132_gettrace/testbench/tb_top.v" Line 221
```

mycpu

```
Test begin!
----[  14045 ns] Number 8'd01 Functional Test Point PASS!!!
----[  41495 ns] Number 8'd02 Functional Test Point PASS!!!
----[  62925 ns] Number 8'd03 Functional Test Point PASS!!!
----[  90375 ns] Number 8'd04 Functional Test Point PASS!!!
----[ 117825 ns] Number 8'd05 Functional Test Point PASS!!!
----[ 145275 ns] Number 8'd06 Functional Test Point PASS!!!
----[ 172725 ns] Number 8'd07 Functional Test Point PASS!!!
----[ 200175 ns] Number 8'd08 Functional Test Point PASS!!!
----[ 227625 ns] Number 8'd09 Functional Test Point PASS!!!
----[ 255075 ns] Number 8'd10 Functional Test Point PASS!!!
----[ 276505 ns] Number 8'd11 Functional Test Point PASS!!!
----[ 297935 ns] Number 8'd12 Functional Test Point PASS!!!
----[ 319365 ns] Number 8'd13 Functional Test Point PASS!!!
----[ 330605 ns] Number 8'd14 Functional Test Point PASS!!!
----[ 342695 ns] Number 8'd15 Functional Test Point PASS!!!
----[ 407295 ns] Number 8'd16 Functional Test Point PASS!!!
----[ 475855 ns] Number 8'd17 Functional Test Point PASS!!!
----[ 532215 ns] Number 8'd18 Functional Test Point PASS!!!
----[ 580575 ns] Number 8'd19 Functional Test Point PASS!!!
----[ 582455 ns] Number 8'd20 Functional Test Point PASS!!!
----[ 584335 ns] Number 8'd21 Functional Test Point PASS!!!
----[ 585745 ns] Number 8'd22 Functional Test Point PASS!!!
----[ 587165 ns] Number 8'd23 Functional Test Point PASS!!!
----[ 610025 ns] Number 8'd24 Functional Test Point PASS!!!
----[ 631455 ns] Number 8'd25 Functional Test Point PASS!!!
----[ 653865 ns] Number 8'd26 Functional Test Point PASS!!!
----[ 675295 ns] Number 8'd27 Functional Test Point PASS!!!
----[ 696725 ns] Number 8'd28 Functional Test Point PASS!!!
----[ 718155 ns] Number 8'd29 Functional Test Point PASS!!!
----[ 739585 ns] Number 8'd30 Functional Test Point PASS!!!
----[ 761015 ns] Number 8'd31 Functional Test Point PASS!!!
----[ 788465 ns] Number 8'd32 Functional Test Point PASS!!!
----[ 815915 ns] Number 8'd33 Functional Test Point PASS!!!
----[ 843365 ns] Number 8'd34 Functional Test Point PASS!!!
----[ 905885 ns] Number 8'd35 Functional Test Point PASS!!!
----[ 993325 ns] Number 8'd36 Functional Test Point PASS!!!
----[1020055 ns] Number 8'd37 Functional Test Point PASS!!!
----[1044205 ns] Number 8'd38 Functional Test Point PASS!!!
----[1053745 ns] Number 8'd39 Functional Test Point PASS!!!
----[1063265 ns] Number 8'd40 Functional Test Point PASS!!!
----[1072165 ns] Number 8'd41 Functional Test Point PASS!!!
----[1077305 ns] Number 8'd42 Functional Test Point PASS!!!
----[1101865 ns] Number 8'd43 Functional Test Point PASS!!!
----[1140075 ns] Number 8'd44 Functional Test Point PASS!!!
----[1178045 ns] Number 8'd45 Functional Test Point PASS!!!
----[1216255 ns] Number 8'd46 Functional Test Point PASS!!!
----[1245945 ns] Number 8'd47 Functional Test Point PASS!!!
----[1290205 ns] Number 8'd48 Functional Test Point PASS!!!
----[1336495 ns] Number 8'd49 Functional Test Point PASS!!!
----[1387255 ns] Number 8'd50 Functional Test Point PASS!!!
----[1388925 ns] Number 8'd51 Functional Test Point PASS!!!
----[1395085 ns] Number 8'd52 Functional Test Point PASS!!!
----[1401885 ns] Number 8'd53 Functional Test Point PASS!!!
----[1408365 ns] Number 8'd54 Functional Test Point PASS!!!
----[1415165 ns] Number 8'd55 Functional Test Point PASS!!!
----[1422595 ns] Number 8'd56 Functional Test Point PASS!!!
----[1429385 ns] Number 8'd57 Functional Test Point PASS!!!
----[1434365 ns] Number 8'd58 Functional Test Point PASS!!!
----[1460075 ns] Number 8'd59 Functional Test Point PASS!!!
----[1487665 ns] Number 8'd60 Functional Test Point PASS!!!
----[1507925 ns] Number 8'd61 Functional Test Point PASS!!!
----[1531075 ns] Number 8'd62 Functional Test Point PASS!!!
----[1544165 ns] Number 8'd63 Functional Test Point PASS!!!
----[1558035 ns] Number 8'd64 Functional Test Point PASS!!!
----[1587035 ns] Number 8'd65 Functional Test Point PASS!!!
----[1614255 ns] Number 8'd66 Functional Test Point PASS!!!
----[1629675 ns] Number 8'd67 Functional Test Point PASS!!!
----[1644675 ns] Number 8'd68 Functional Test Point PASS!!!
----[1649965 ns] Number 8'd69 Functional Test Point PASS!!!
============================================================
Test end!
----PASS!!!
$finish called at time : 1650425 ns : File "D:/Architecture/lab1/
↪  lab3/CPU_CDE/mycpu_verify/testbench/mycpu_tb.v" Line 267
```

`func_lab9` 两边都到第 94 点。`gettrace` 在 2131085 ns 结束，`mycpu` 在 1940275 ns 结束。第 69 点仍是 1841065 ns 与 1649965 ns。从第 42 点起两边相

差约 0.116 ms，到第 68 点及以后约为 0.191 ms，与实验 2 里乘法之后拉开的差距相同。前 69 点与上一节相同，下面分成两段。

gettrace

```
Test begin!
----[  14055 ns] Number 8'd01 Functional Test Point PASS!!!
----[  41545 ns] Number 8'd02 Functional Test Point PASS!!!
----[  63015 ns] Number 8'd03 Functional Test Point PASS!!!
----[  90505 ns] Number 8'd04 Functional Test Point PASS!!!
----[ 117995 ns] Number 8'd05 Functional Test Point PASS!!!
----[ 145485 ns] Number 8'd06 Functional Test Point PASS!!!
----[ 172975 ns] Number 8'd07 Functional Test Point PASS!!!
----[ 200465 ns] Number 8'd08 Functional Test Point PASS!!!
----[ 227955 ns] Number 8'd09 Functional Test Point PASS!!!
----[ 255445 ns] Number 8'd10 Functional Test Point PASS!!!
----[ 276915 ns] Number 8'd11 Functional Test Point PASS!!!
----[ 298385 ns] Number 8'd12 Functional Test Point PASS!!!
----[ 319855 ns] Number 8'd13 Functional Test Point PASS!!!
----[ 333695 ns] Number 8'd14 Functional Test Point PASS!!!
----[ 348585 ns] Number 8'd15 Functional Test Point PASS!!!
----[ 420255 ns] Number 8'd16 Functional Test Point PASS!!!
----[ 496875 ns] Number 8'd17 Functional Test Point PASS!!!
----[ 559275 ns] Number 8'd18 Functional Test Point PASS!!!
----[ 613675 ns] Number 8'd19 Functional Test Point PASS!!!
----[ 615745 ns] Number 8'd20 Functional Test Point PASS!!!
----[ 617815 ns] Number 8'd21 Functional Test Point PASS!!!
----[ 619425 ns] Number 8'd22 Functional Test Point PASS!!!
----[ 621035 ns] Number 8'd23 Functional Test Point PASS!!!
----[ 643935 ns] Number 8'd24 Functional Test Point PASS!!!
----[ 665405 ns] Number 8'd25 Functional Test Point PASS!!!
----[ 687855 ns] Number 8'd26 Functional Test Point PASS!!!
----[ 709325 ns] Number 8'd27 Functional Test Point PASS!!!
----[ 730795 ns] Number 8'd28 Functional Test Point PASS!!!
----[ 752265 ns] Number 8'd29 Functional Test Point PASS!!!
----[ 773735 ns] Number 8'd30 Functional Test Point PASS!!!
----[ 795205 ns] Number 8'd31 Functional Test Point PASS!!!
----[ 822695 ns] Number 8'd32 Functional Test Point PASS!!!
----[ 850185 ns] Number 8'd33 Functional Test Point PASS!!!
----[ 877675 ns] Number 8'd34 Functional Test Point PASS!!!
----[ 935355 ns] Number 8'd35 Functional Test Point PASS!!!
----[1019275 ns] Number 8'd36 Functional Test Point PASS!!!
----[1093745 ns] Number 8'd37 Functional Test Point PASS!!!
----[1160535 ns] Number 8'd38 Functional Test Point PASS!!!
----[1170115 ns] Number 8'd39 Functional Test Point PASS!!!
----[1179675 ns] Number 8'd40 Functional Test Point PASS!!!
----[1188615 ns] Number 8'd41 Functional Test Point PASS!!!
----[1193795 ns] Number 8'd42 Functional Test Point PASS!!!
----[1222025 ns] Number 8'd43 Functional Test Point PASS!!!
----[1265245 ns] Number 8'd44 Functional Test Point PASS!!!
----[1308185 ns] Number 8'd45 Functional Test Point PASS!!!
----[1351405 ns] Number 8'd46 Functional Test Point PASS!!!
----[1385005 ns] Number 8'd47 Functional Test Point PASS!!!
----[1432805 ns] Number 8'd48 Functional Test Point PASS!!!
----[1482775 ns] Number 8'd49 Functional Test Point PASS!!!
----[1537775 ns] Number 8'd50 Functional Test Point PASS!!!
----[1539745 ns] Number 8'd51 Functional Test Point PASS!!!
----[1547845 ns] Number 8'd52 Functional Test Point PASS!!!
----[1556585 ns] Number 8'd53 Functional Test Point PASS!!!
----[1565005 ns] Number 8'd54 Functional Test Point PASS!!!
----[1573745 ns] Number 8'd55 Functional Test Point PASS!!!
----[1583125 ns] Number 8'd56 Functional Test Point PASS!!!
----[1591865 ns] Number 8'd57 Functional Test Point PASS!!!
----[1598255 ns] Number 8'd58 Functional Test Point PASS!!!
----[1627845 ns] Number 8'd59 Functional Test Point PASS!!!
----[1659605 ns] Number 8'd60 Functional Test Point PASS!!!
----[1682825 ns] Number 8'd61 Functional Test Point PASS!!!
----[1709545 ns] Number 8'd62 Functional Test Point PASS!!!
----[1724385 ns] Number 8'd63 Functional Test Point PASS!!!
----[1740065 ns] Number 8'd64 Functional Test Point PASS!!!
----[1771905 ns] Number 8'd65 Functional Test Point PASS!!!
----[1801825 ns] Number 8'd66 Functional Test Point PASS!!!
----[1818865 ns] Number 8'd67 Functional Test Point PASS!!!
----[1835425 ns] Number 8'd68 Functional Test Point PASS!!!
----[1841065 ns] Number 8'd69 Functional Test Point PASS!!!
```

mycpu

```
Test begin!
----[  14045 ns] Number 8'd01 Functional Test Point PASS!!!
----[  41495 ns] Number 8'd02 Functional Test Point PASS!!!
----[  62925 ns] Number 8'd03 Functional Test Point PASS!!!
----[  90375 ns] Number 8'd04 Functional Test Point PASS!!!
----[ 117825 ns] Number 8'd05 Functional Test Point PASS!!!
----[ 145275 ns] Number 8'd06 Functional Test Point PASS!!!
----[ 172725 ns] Number 8'd07 Functional Test Point PASS!!!
----[ 200175 ns] Number 8'd08 Functional Test Point PASS!!!
----[ 227625 ns] Number 8'd09 Functional Test Point PASS!!!
----[ 255075 ns] Number 8'd10 Functional Test Point PASS!!!
----[ 276505 ns] Number 8'd11 Functional Test Point PASS!!!
----[ 297935 ns] Number 8'd12 Functional Test Point PASS!!!
----[ 319365 ns] Number 8'd13 Functional Test Point PASS!!!
----[ 330605 ns] Number 8'd14 Functional Test Point PASS!!!
----[ 342695 ns] Number 8'd15 Functional Test Point PASS!!!
----[ 407295 ns] Number 8'd16 Functional Test Point PASS!!!
----[ 475855 ns] Number 8'd17 Functional Test Point PASS!!!
----[ 532215 ns] Number 8'd18 Functional Test Point PASS!!!
----[ 580575 ns] Number 8'd19 Functional Test Point PASS!!!
----[ 582455 ns] Number 8'd20 Functional Test Point PASS!!!
----[ 584335 ns] Number 8'd21 Functional Test Point PASS!!!
----[ 585745 ns] Number 8'd22 Functional Test Point PASS!!!
----[ 587165 ns] Number 8'd23 Functional Test Point PASS!!!
----[ 610025 ns] Number 8'd24 Functional Test Point PASS!!!
----[ 631455 ns] Number 8'd25 Functional Test Point PASS!!!
----[ 653865 ns] Number 8'd26 Functional Test Point PASS!!!
----[ 675295 ns] Number 8'd27 Functional Test Point PASS!!!
----[ 696725 ns] Number 8'd28 Functional Test Point PASS!!!
----[ 718155 ns] Number 8'd29 Functional Test Point PASS!!!
----[ 739585 ns] Number 8'd30 Functional Test Point PASS!!!
----[ 761015 ns] Number 8'd31 Functional Test Point PASS!!!
----[ 788465 ns] Number 8'd32 Functional Test Point PASS!!!
----[ 815915 ns] Number 8'd33 Functional Test Point PASS!!!
----[ 843365 ns] Number 8'd34 Functional Test Point PASS!!!
----[ 905885 ns] Number 8'd35 Functional Test Point PASS!!!
----[ 993325 ns] Number 8'd36 Functional Test Point PASS!!!
----[1020055 ns] Number 8'd37 Functional Test Point PASS!!!
----[1044205 ns] Number 8'd38 Functional Test Point PASS!!!
----[1053745 ns] Number 8'd39 Functional Test Point PASS!!!
----[1063265 ns] Number 8'd40 Functional Test Point PASS!!!
----[1072165 ns] Number 8'd41 Functional Test Point PASS!!!
----[1077305 ns] Number 8'd42 Functional Test Point PASS!!!
----[1101865 ns] Number 8'd43 Functional Test Point PASS!!!
----[1140075 ns] Number 8'd44 Functional Test Point PASS!!!
----[1178045 ns] Number 8'd45 Functional Test Point PASS!!!
----[1216255 ns] Number 8'd46 Functional Test Point PASS!!!
----[1245945 ns] Number 8'd47 Functional Test Point PASS!!!
----[1290205 ns] Number 8'd48 Functional Test Point PASS!!!
----[1336495 ns] Number 8'd49 Functional Test Point PASS!!!
----[1387255 ns] Number 8'd50 Functional Test Point PASS!!!
----[1388925 ns] Number 8'd51 Functional Test Point PASS!!!
----[1395085 ns] Number 8'd52 Functional Test Point PASS!!!
----[1401885 ns] Number 8'd53 Functional Test Point PASS!!!
----[1408365 ns] Number 8'd54 Functional Test Point PASS!!!
----[1415165 ns] Number 8'd55 Functional Test Point PASS!!!
----[1422595 ns] Number 8'd56 Functional Test Point PASS!!!
----[1429385 ns] Number 8'd57 Functional Test Point PASS!!!
----[1434365 ns] Number 8'd58 Functional Test Point PASS!!!
----[1460075 ns] Number 8'd59 Functional Test Point PASS!!!
----[1487665 ns] Number 8'd60 Functional Test Point PASS!!!
----[1507925 ns] Number 8'd61 Functional Test Point PASS!!!
----[1531075 ns] Number 8'd62 Functional Test Point PASS!!!
----[1544165 ns] Number 8'd63 Functional Test Point PASS!!!
----[1558035 ns] Number 8'd64 Functional Test Point PASS!!!
----[1587035 ns] Number 8'd65 Functional Test Point PASS!!!
----[1614255 ns] Number 8'd66 Functional Test Point PASS!!!
----[1629675 ns] Number 8'd67 Functional Test Point PASS!!!
----[1644675 ns] Number 8'd68 Functional Test Point PASS!!!
----[1649965 ns] Number 8'd69 Functional Test Point PASS!!!
```

gettrace

```
----[1846915 ns] Number 8'd70 Functional Test Point PASS!!!
----[1853605 ns] Number 8'd71 Functional Test Point PASS!!!
----[1860205 ns] Number 8'd72 Functional Test Point PASS!!!
----[1866895 ns] Number 8'd73 Functional Test Point PASS!!!
----[1874275 ns] Number 8'd74 Functional Test Point PASS!!!
----[1881675 ns] Number 8'd75 Functional Test Point PASS!!!
----[1889055 ns] Number 8'd76 Functional Test Point PASS!!!
----[1896725 ns] Number 8'd77 Functional Test Point PASS!!!
----[1904395 ns] Number 8'd78 Functional Test Point PASS!!!
----[1911965 ns] Number 8'd79 Functional Test Point PASS!!!
----[1918905 ns] Number 8'd80 Functional Test Point PASS!!!
----[1937725 ns] Number 8'd81 Functional Test Point PASS!!!
----[1946945 ns] Number 8'd82 Functional Test Point PASS!!!
----[1962245 ns] Number 8'd83 Functional Test Point PASS!!!
----[1977545 ns] Number 8'd84 Functional Test Point PASS!!!
----[1992845 ns] Number 8'd85 Functional Test Point PASS!!!
----[2008145 ns] Number 8'd86 Functional Test Point PASS!!!
----[2023445 ns] Number 8'd87 Functional Test Point PASS!!!
----[2038745 ns] Number 8'd88 Functional Test Point PASS!!!
----[2054055 ns] Number 8'd89 Functional Test Point PASS!!!
----[2069375 ns] Number 8'd90 Functional Test Point PASS!!!
----[2084685 ns] Number 8'd91 Functional Test Point PASS!!!
----[2099995 ns] Number 8'd92 Functional Test Point PASS!!!
----[2115305 ns] Number 8'd93 Functional Test Point PASS!!!
----[2130615 ns] Number 8'd94 Functional Test Point PASS!!!
================================================================
gettrace end!
----Succeed in generating trace file!
$finish called at time : 2131085 ns : File "D:/Architecture/lab1/
↪  lab3/CPU_CDE/cpu132_gettrace/testbench/tb_top.v" Line 221
```

mycpu

```
----[1655465 ns] Number 8'd70 Functional Test Point PASS!!!
----[1661805 ns] Number 8'd71 Functional Test Point PASS!!!
----[1668055 ns] Number 8'd72 Functional Test Point PASS!!!
----[1674395 ns] Number 8'd73 Functional Test Point PASS!!!
----[1681565 ns] Number 8'd74 Functional Test Point PASS!!!
----[1688755 ns] Number 8'd75 Functional Test Point PASS!!!
----[1695925 ns] Number 8'd76 Functional Test Point PASS!!!
----[1703385 ns] Number 8'd77 Functional Test Point PASS!!!
----[1710845 ns] Number 8'd78 Functional Test Point PASS!!!
----[1718215 ns] Number 8'd79 Functional Test Point PASS!!!
----[1724805 ns] Number 8'd80 Functional Test Point PASS!!!
----[1742565 ns] Number 8'd81 Functional Test Point PASS!!!
----[1750735 ns] Number 8'd82 Functional Test Point PASS!!!
----[1766485 ns] Number 8'd83 Functional Test Point PASS!!!
----[1782235 ns] Number 8'd84 Functional Test Point PASS!!!
----[1797985 ns] Number 8'd85 Functional Test Point PASS!!!
----[1813735 ns] Number 8'd86 Functional Test Point PASS!!!
----[1829485 ns] Number 8'd87 Functional Test Point PASS!!!
----[1845235 ns] Number 8'd88 Functional Test Point PASS!!!
----[1860995 ns] Number 8'd89 Functional Test Point PASS!!!
----[1876765 ns] Number 8'd90 Functional Test Point PASS!!!
----[1892525 ns] Number 8'd91 Functional Test Point PASS!!!
----[1908285 ns] Number 8'd92 Functional Test Point PASS!!!
----[1924045 ns] Number 8'd93 Functional Test Point PASS!!!
----[1939805 ns] Number 8'd94 Functional Test Point PASS!!!
================================================================
Test end!
----PASS!!!
$finish called at time : 1940275 ns : File "D:/Architecture/lab1/
↪  lab3/CPU_CDE/mycpu_verify/testbench/mycpu_tb.v" Line 267
```

程序	mycpu 结束	gettrace 结束
func_lab8（69 点）	1.650 ms	1.842 ms
func_lab9（94 点）	1.940 ms	2.131 ms

6 实验收获

例外和中断都要到写回级才改 CP0，更年轻的指令在例外进入执行级时就被丢掉，这样 SYSCALL 后面的存储不会先于处理程序写进数据 RAM。MFC0 的结果出现得比 ALU 晚，处理程序里紧接着的比较要按 load-use 再停两级。Cause、Status 的位必须和参考 CPU 一致，因为 trace 比较的是 MFC0 写出的整字，处理程序自己的掩码盖不住多余的 TI 或 IP。Count 隔拍加 1、IP7 比 TI 晚一拍之后，定时器和软件中断都能在自旋分支上被受理，EPC 落在该分支上，处理程序加 8 后跳过分支和它的延迟槽。

7 附：源码目录

CPU 仍在实验 1 的工程里改。lab3 放测试程序、仿真脚本和本报告。

- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/cp0.v
- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/IF_stage.v
- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/ID_stage.v
- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/EXE_stage.v
- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/MEM_stage.v

- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/WB_stage.v
- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/mycpu.h
- lab1/lab3/CPU_CDE/mycpu_verify/rtl/myCPU/mycpu_top.v
- lab1/lab3/CPU_CDE/cpu132_gettrace/golden_trace.txt
- lab1/lab3/CPU_CDE/soft/func_lab8/obj/inst_ram.coe
- lab1/lab3/CPU_CDE/soft/func_lab9/obj/inst_ram.coe
- lab3/sim/run_gettrace_lab8.tcl
- lab3/sim/run_mycpu_lab8.tcl
- lab3/sim/run_gettrace_lab9.tcl
- lab3/sim/run_mycpu_lab9.tcl
- lab3/report/report.tex